

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q18-0933

San Bruno Tap 138 kV  
May 25, 2019

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0933. The Interconnection Customer proposes a 63.2 MW solar photovoltaic facility interconnecting at the San Bruno Tap 138 kV substation in the West Virginia area. The request was received on January 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$61,126,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Sherbino I / Coal 138 KV line  
Contingency: HintonAp / Antelope Vly Station 138 KV  
Loading: 165.5% (Rating: 233 MVA)  
DFAX: 7.5%  
Pre-project loading: 163.5%
2. Facility: S Windsor / Dexter Tap 138 KV  
Contingency: Hinton Ap / Antelope Vly Station 138 KV  
Loading: 153.6% (Rating: 154 MVA)  
DFAX: 7.5%  
Pre-project loading: 150.5%
3. Facility: S Windsor / Tilton 138 KV line  
Contingency: Hnton Ap / Antelope Valley Station 138 KV  
Loading: 143.1% (Rating: 166 MVA)  
DFAX: 7.5%  
Pre-project loading: 140.2%
4. Facility: Revere Co / Ormond 138 KV  
Contingency: Davidson / Lore 138 KV  
Loading: 131.8% (Rating: 253 MVA)  
DFAX: 26.0%  
Pre-project loading: 125.4%
5. Facility: Lore / Davidson 138 KV  
Contingency: Ormond Substation / Revere Co 138 KV  
Loading: 125.7% (Rating: 262 MVA)  
DFAX: 20.2%  
Pre-project loading: 120.8%

6. Facility: Davidson Substation / Whites Creek 138 KV  
Contingency: Revere o / Ormnd 138 KV line  
Loading: 125.4% (Rating: 260 MVA)  
DFAX: 20.2%  
Pre-project loading: 120.5%
7. Facility: Dextr Tap / Berlin Center 138 KV line  
Contingency: Hinton Ap / Antelope Valley Station 138 KV  
Loading: 124.3% (Rating: 183 MVA)  
DFAX: 7.5%  
Pre-project loading: 121.7%
8. Facility: McClain Energy Facility / Coal 138 KV  
Contingency: Manatee / Hinton Ap 345 KV Line  
Loading: 114.8% (Rating: 374 MVA)  
DFAX: 5.1%  
Pre-project loading: 113.9%
9. Facility: Beardsley St / Neosho 138 KV  
Contingency: Ormond / San Bruno Tap 138 KV  
Loading: 114.7% (Rating: 249 MVA)  
DFAX: 5.2%  
Pre-project loading: 113.3%
10. Facility: Ormond / San Bruno Tap 138 KV  
Contingency: Lore 345 kV/138 kV XFMR  
Loading: 111.1% (Rating: 175 MVA)  
DFAX: 61.6%  
Pre-project loading: 88.9%
11. Facility: Browning / Willmar 138 KV  
Contingency: Whites Creek / Sedalia 138 KV  
Loading: 110.6% (Rating: 169 MVA)  
DFAX: 16.2%  
Pre-project loading: 104.6%
12. Facility: Colton / Coal 138 KV  
Contingency: McClain Energy Facility / Coal 138 KV line  
Loading: 110.0% (Rating: 233 MVA)  
DFAX: 6.0%  
Pre-project loading: 108.4%
13. Facility: Central Juncos 3202 / Colton 138 KV  
Contingency: McClain Energy Facility / Coal 138 KV  
Loading: 109.2% (Rating: 233 MVA)  
DFAX: 6.0%  
Pre-project loading: 107.5%
14. Facility: Hinton Ap 345-138 kV TX  
Contingency: Manatee / Hinton Ap 345 KV  
Loading: 108.9% (Rating: 640 MVA)  
DFAX: 6.5%  
Pre-project loading: 108.2%
15. Facility: Browning / East Calvert 138 KV line  
Contingency: Sedalia / Whites Creek 138 KV  
Loading: 106.4% (Rating: 171 MVA)  
DFAX: 16.2%  
Pre-project loading: 100.4%
16. Facility: Rawson Packaging / Central Juncos 3202 138 KV  
Contingency: McClain Energy Facility / Coal 138 KV

Loading: 106.2% (Rating: 236 MVA)

DFAX: 6.0%

Pre-project loading: 104.6%

17. Facility: Hancock / Ash Street 138 KV

Contingency: Ormond / San Bruno Tap 138 KV line

Loading: 105.4% (Rating: 232 MVA)

DFAX: 5.2%

Pre-project loading: 104.0%

18. Facility: Huntersville Delivery No 2 / Willar 138 KV

Contingency: Sedalia / Whites rk 138 KV

Loading: 101.5% (Rating: 189 MVA)

DFAX: 16.2%

Pre-project loading: 96.1%

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Sherbino I / Coal 138 KV line                | Rebuild line with higher-capacity conductor | \$2,703,000   | 5% (\$129,000)        |
| S Windsor / Dexter Tap 138 KV                | Rebuild line with higher-capacity conductor | \$9,286,000   | 5% (\$445,000)        |
| S Windsor / Tilton 138 KV line               | Rebuild line with higher-capacity conductor | \$36,265,000  | 5% (\$1,736,000)      |
| Revere Co / Ormond 138 KV                    | Rebuild line with higher-capacity conductor | \$8,936,000   | 100% (\$8,936,000)    |
| Lore / Davidson 138 KV                       | Rebuild line with higher-capacity conductor | \$3,749,000   | 100% (\$3,749,000)    |
| Davidson Substation / Whites Crk 138 KV Line | Rebuild line with higher-capacity conductor | \$4,769,000   | 100% (\$4,769,000)    |
| Dexter Tap / Berlin Center 138 KV            | Rebuild line with higher-capacity conductor | \$10,508,000  | 5% (\$503,000)        |
| Coal / McClain Energy Facility 138 KV line   | Rebuild line with higher-capacity conductor | \$12,716,000  | 6% (\$810,000)        |
| Beardsley St / Neosho 138 KV                 | Rebuild line with higher-capacity conductor | \$9,191,000   | 100% (\$9,191,000)    |
| San Bruno Tap / Ormond 138 KV                | Rebuild line with higher-capacity conductor | \$11,519,000  | 100% (\$11,519,000)   |
| Browning / Willmar 138 KV                    | Rebuild line with higher-capacity conductor | \$3,111,000   | 100% (\$3,111,000)    |
| Colton / Coal Substation 138 KV              | Rebuild line with higher-capacity conductor | \$10,843,000  | 100% (\$10,843,000)   |
| Central Juncos 3202 / Colton 138 KV          | Reconductor line                            | \$1,801,000   | 100% (\$1,801,000)    |

| Facility                                      | Required Upgrade                           | Cost Estimate | Allocation to Project |
|-----------------------------------------------|--------------------------------------------|---------------|-----------------------|
| Hinton Ap 345-138 kV TX                       | Replace transformer with higher-rated unit | \$17,649,000  | 1% (\$107,000)        |
| Browning / East Calvert 138 KV line           | Reconductor line                           | \$741,000     | 100% (\$741,000)      |
| Rawson Packaging / Central Juncos 3202 138 KV | Reconductor line                           | \$1,137,000   | 100% (\$1,137,000)    |
| Hancock / Ash St 138 KV L                     | Reconductor line                           | \$778,000     | 100% (\$778,000)      |
| Huntersville Delivery No 2 / Willar 138 KV    | Reconductor line                           | \$821,000     | 100% (\$821,000)      |
|                                               | Total allocated to project                 |               | \$61,126,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.